MEDICAL MONITORING AND ADVANCED SENSOR DATA PROCESSING

TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

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7-2-2025

1 TASKS

Q1 Consider the data "Stairs". Resample the smartwatch signals to 1000 Hz and synchronize all signals using the artifact you created at the beginning of your measurement. Cut the signals in order that it only contains the exercise. Plot the PPG (biopac), PPG (smartwatch), ECG (biopac), Acceleration data (smartwatch) each in an own plot (4 plots) with corresponding time axis. Mark the transition between walking and resting in your plots.

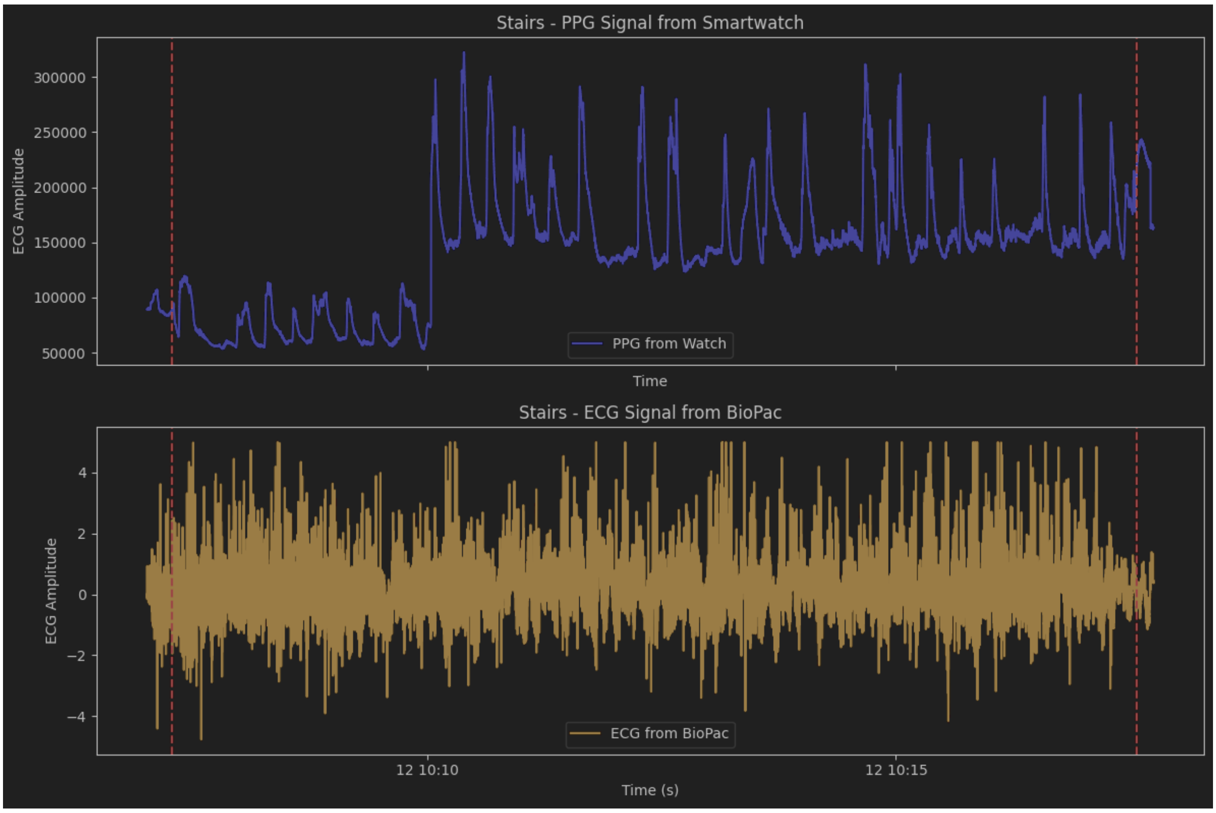


Figure 1: PPG containing only the exercise from Smartwatch (top) and ECG containing the same exercise from BioPac (bottom)

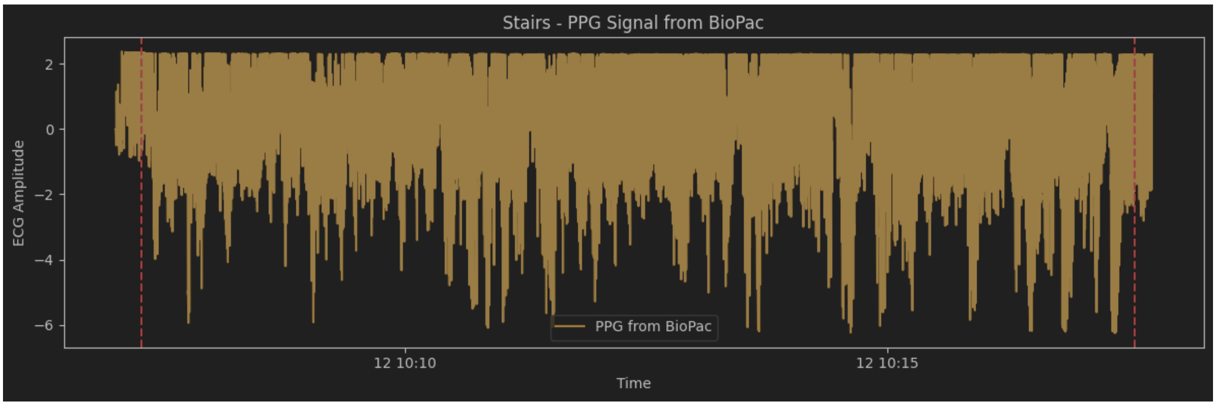


Figure 2: PPG containing only the exercise from BioPac

TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

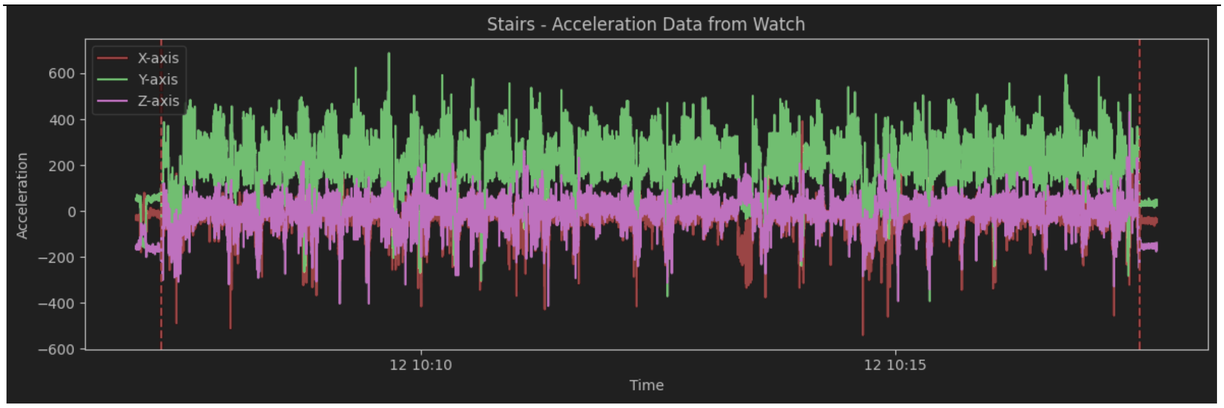


Figure 3: Acceleration Data from Smartwatch during exercise

Q2 Consider the data "Workout". Resample the smartwatch signals to 1000 Hz and synchronize all signals using the artifact you created at the beginning of your measurement. Cut the signals in order that it only contains the exercise. Plot the PPG (biopac), PPG (smartwatch), ECG (biopac), Acceleration data (smartwatch) each in an own plot (4 plots) with corresponding time axis. Mark the transitions between exercise and resting.

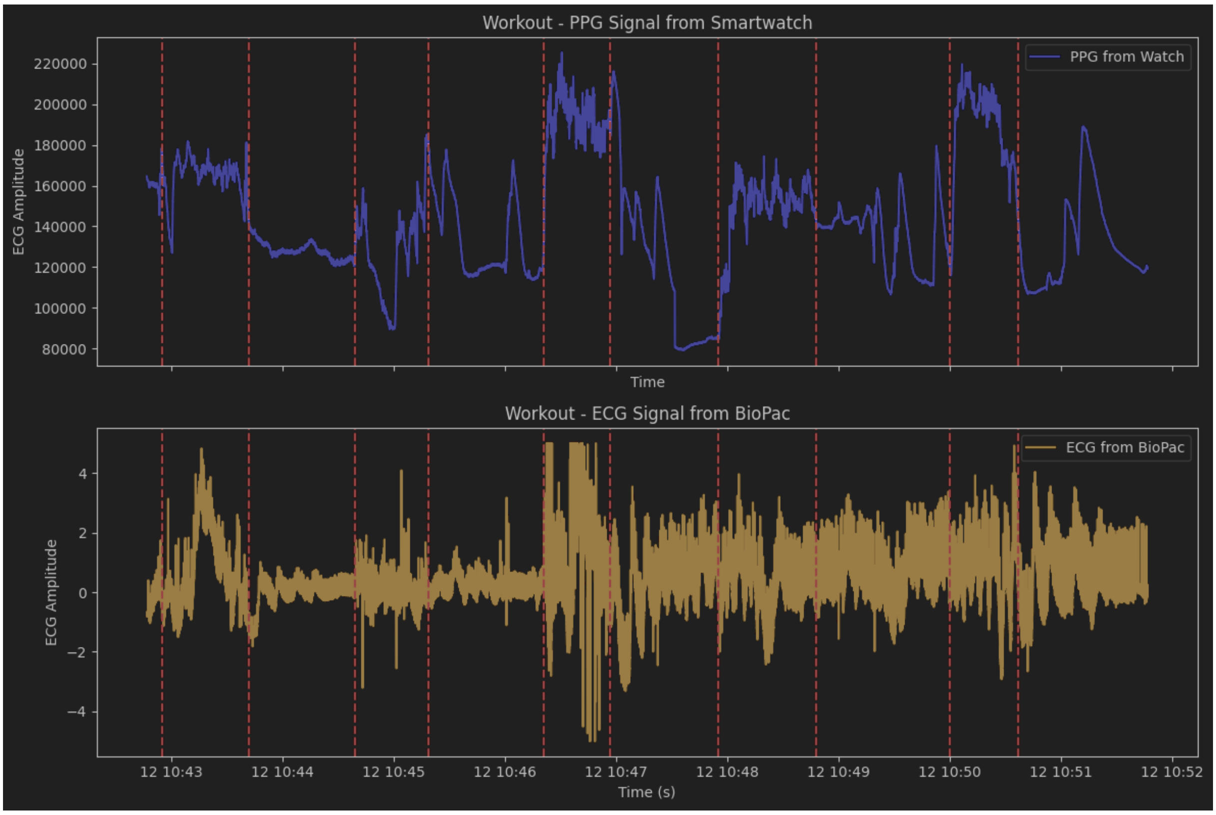


Figure 4: PPG containing only the exercise from Smartwatch (top) and ECG containing the same exercise from BioPac (bottom), with markers for the transitions between exercises and breaks

TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

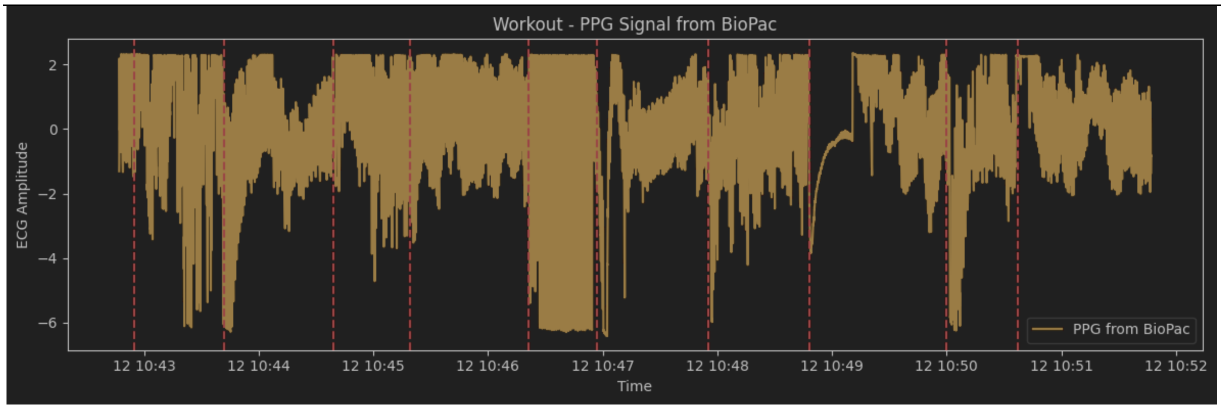


Figure 5: PPG containing only the exercise from BioPac, with markers for the transitions between exercises and breaks

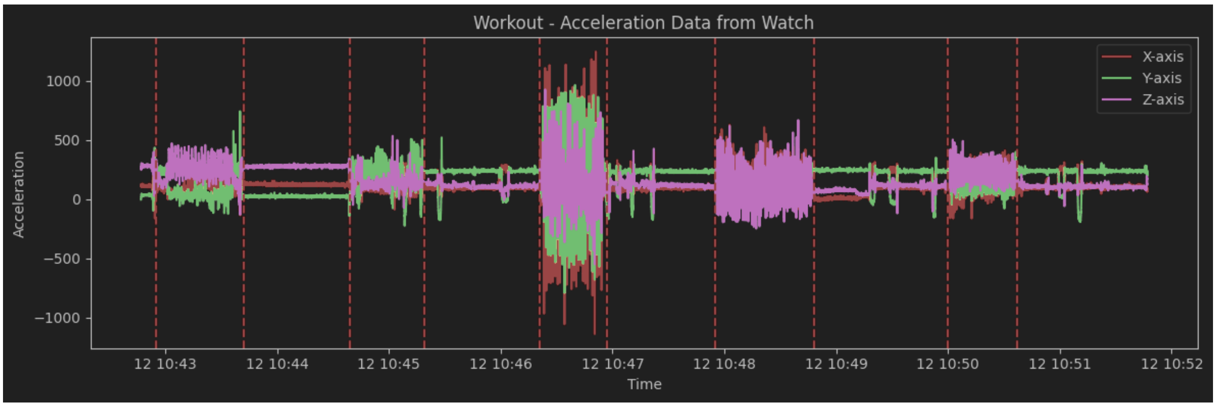


Figure 6: Acceleration Data from Smartwatch during exercise, with markers for the transitions between exercises and breaks

Q3 Consider the data "Stairs". Use the neurokit2 toolbox [3] to perform R-peak detection of the ECG.
Then calculate the instantaneous heart rate (i.e. the heart rate for each detected beat) without (!) the help of the toolbox.
Calculate a smoothed heart rate by taking the median per 10s window sliding it by 1s and assign the calculated value to the middle of the window. That way you have one heart rate per second.
Plot a 20s segment of the ECG with the corresponding R-peak detections, where the detection worked well.
Plot a 20s segment of the ECG with the corresponding R-peak detections, where the detection was faulty.
Plot the course of the instantaneous heart rate over time together with the course of the smoothed heart rate where you used sliding windows.
Discuss differences that occur between the two heart rates (10 sentences max)

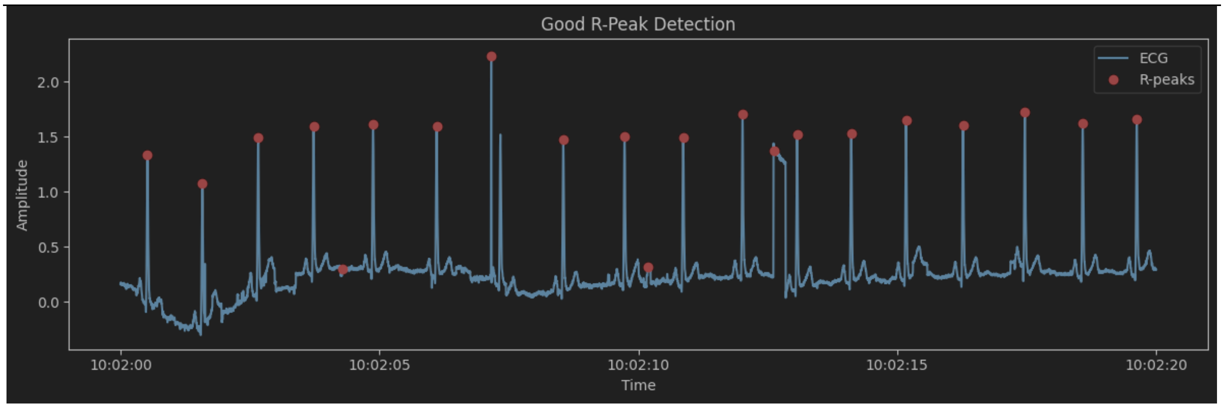


Figure 7: R-Peak Detection that worked well on BioPac ECG

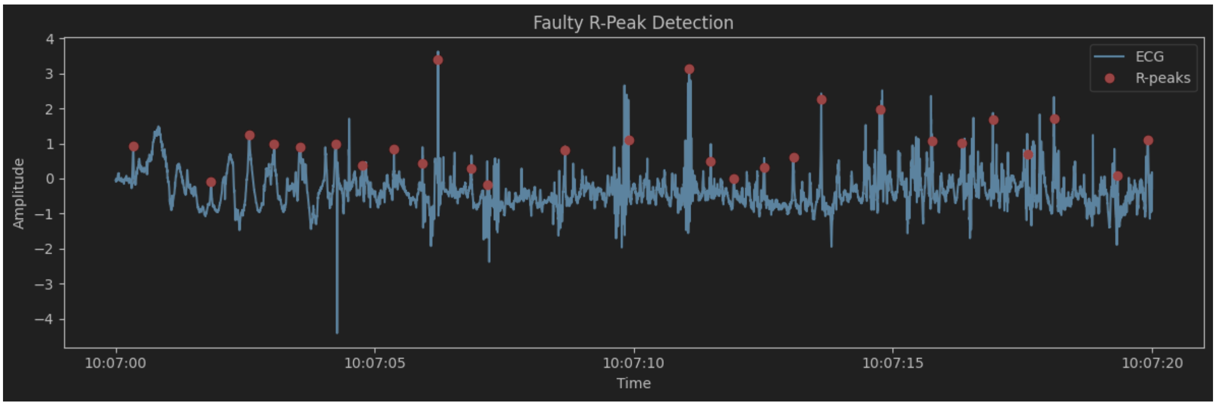


Figure 8: R-Peak Detection that worked poorly on BioPac ECG

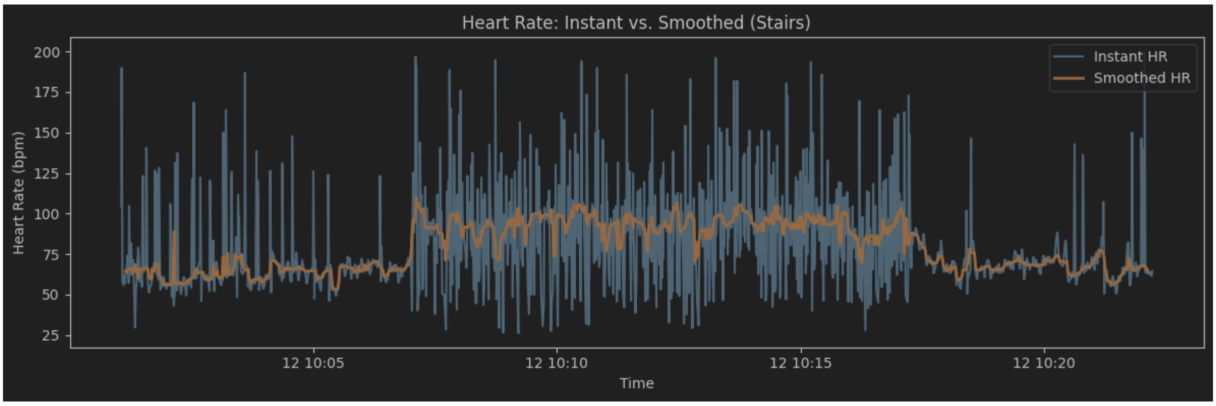


Figure 9: Derived heart rate during the exercise: instant heart rate (blue) and smoothed heart rate (orange)

The instant heart rate shows extremely sharp peaks and declines that realistically do not give meaningful information about the heart rate. For instance, at the beginning of the measurement, before the start of the exercise (during rest), the instant heart rate hits almost 200 for an instance, and drops to 30 just a few heart beats later. This is not in any way descriptive of the heart rate we want to analyze, as those peaks and drops are most likely artefacts of measuring intervals or slight irregularities in the heart rate. If two heart beats are slightly further apart than usual, but only those two heartbeats, the instant heart rate infers from that measurement that the heart rate suddenly dropped to 25 for just that one instance, which is not what happens if you consider a longer time frame. The smoothed heart rate takes the average measurements over a 10s window instead, which yields a much more reasonable representation of the heart rate. As the heart rate describes "the number of heart beats in a minute" (bpm = beats per minute), making assumptions on its value on just one instance is always going to be much further from the truth than a smoothed average, and trying to derive it in real time, beat for beat, will always yield noise above anything else.

Q4 Consider the data "Workout". Use the neurokit2 toolbox [3] to perform R-peak detection of the ECG.
Then calculate the heart rate for each detected beat without(!) the help of the toolbox.
Calculate a smoothed heart rate by taking the mean per 10s window sliding it by 1s and assign the calculated value to the middle of the window. That way you have one heart rate per second.
Plot a 20s segment of the ECG with the corresponding R-peak detections, where the detection worked well.
Plot a 20s segment of the ECG with the corresponding R-peak detections where the detection was faulty.
Plot the course of the instantaneous heart rate over time together with the course of the smoothed heart rate where you used sliding windows.
Do you notice any differences between the two heart rate courses?

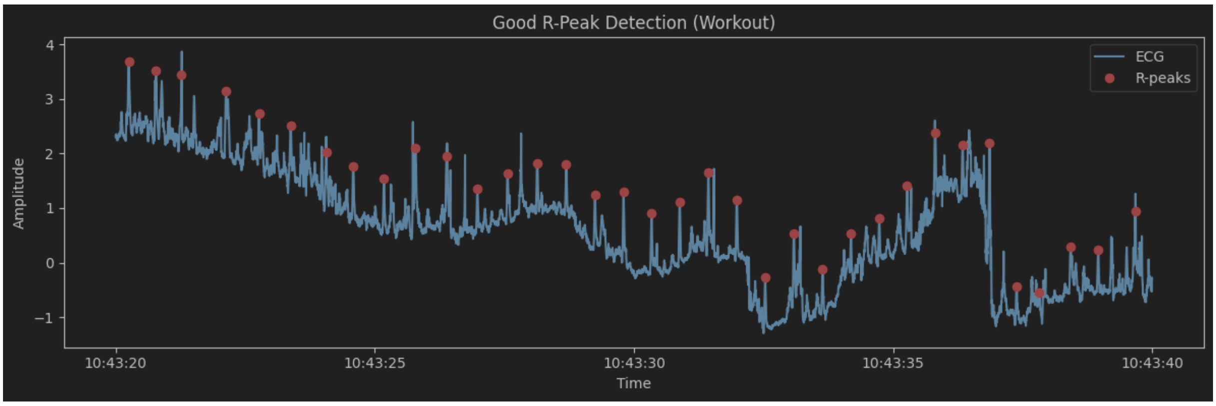


Figure 10: R-Peak Detection that worked well on BioPac ECG

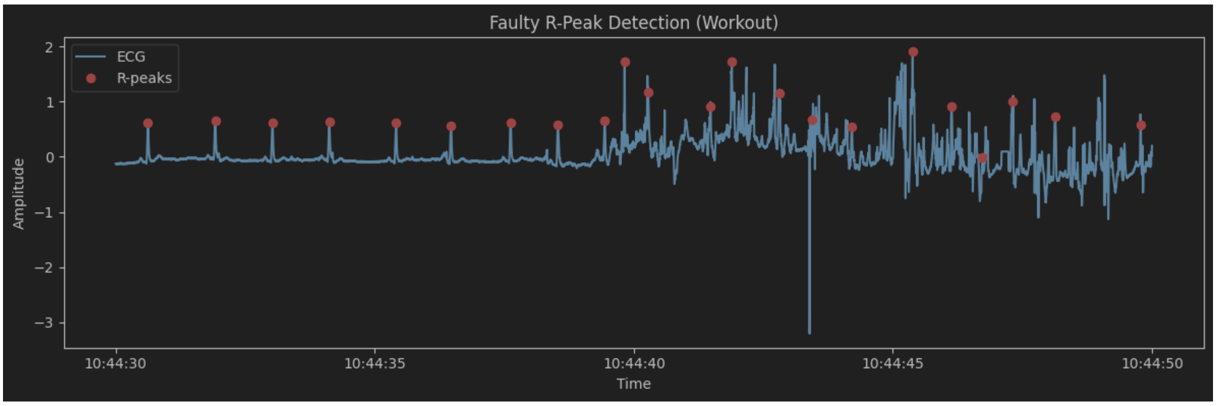


Figure 11: R-Peak Detection that worked poorly on BioPac ECG

TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

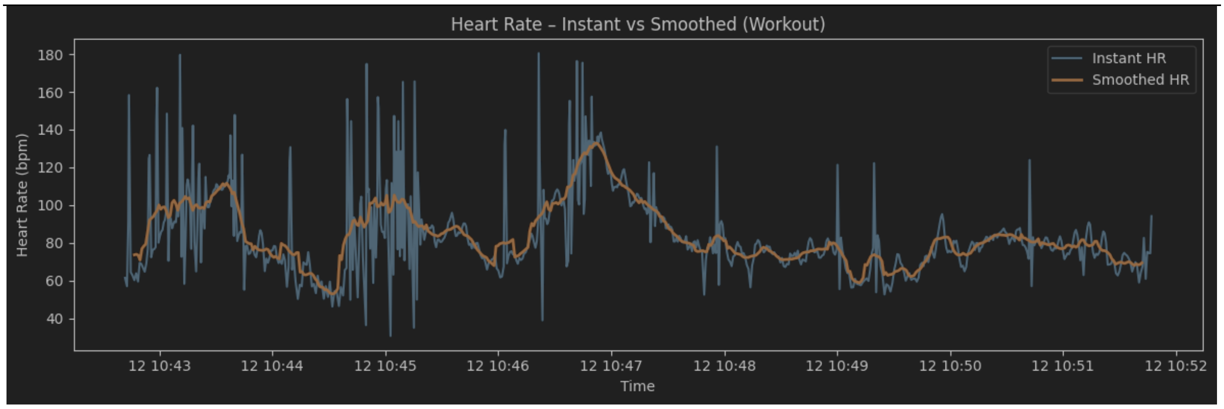


Figure 12: Derived heart rate during the exercise: instant heart rate (blue) and smoothed heart rate (orange)

The smoothed heart rate is more resistant to irregularities and therefore much more readable. It is much easier to track where the subject was doing exercises and where the subject was resting in the smoothed heart rate. Notably, periods of higher smoothed heart rates largely overlap with more sharp spikes and drops in the instant heart rate. The instant heart rate appears more irregular when the subjects heart beast faster.

Q5 Consider the data "Stairs" and use the smartwatch PPG data. Use the neurokit2 toolbox [3] to perform PPG peak detection.
Then calculate the heart rate for each detected beat without(!) the help of the toolbox.
Calculate a smoothed heart rate by taking the mean per 10s window sliding it by 1s and assign the calculated value to the middle of the window. That way you have one heart rate per second starting at 5s.
Plot a 20s segment of the PPG with the corresponding beat detections, where the detection worked well.
Plot a 20s segment of the PPG with the corresponding beat detections, where the detection was faulty.
Plot the course of the instantaneous heart rate over time together with the course of the smoothed heart rate where you used sliding windows.
Do you notice any differences between the two heart rate courses?

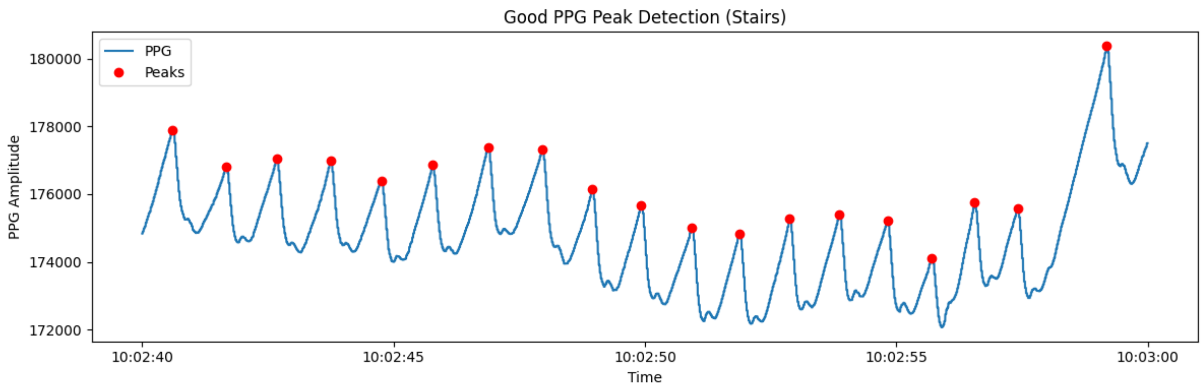


Figure 13: R-Peak Detection that worked well on Smartwatch PPG

TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

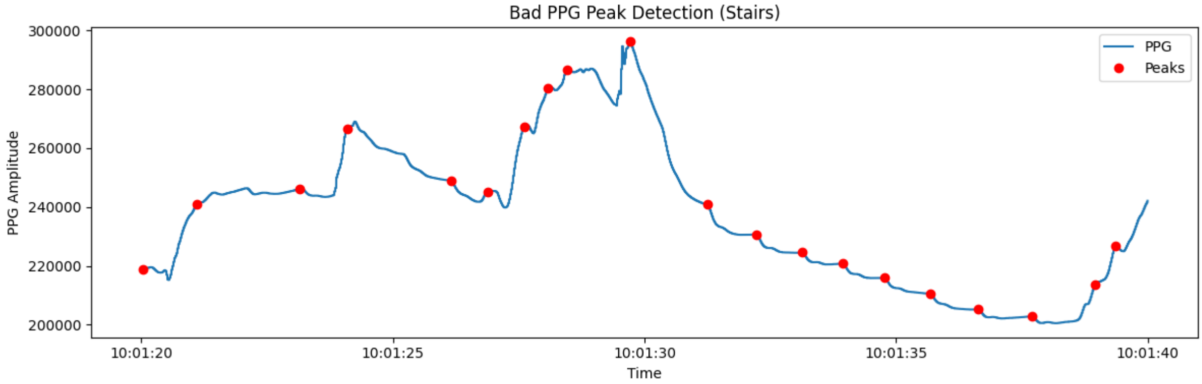


Figure 14: R-Peak Detection that worked poorly on Smartwatch PPG

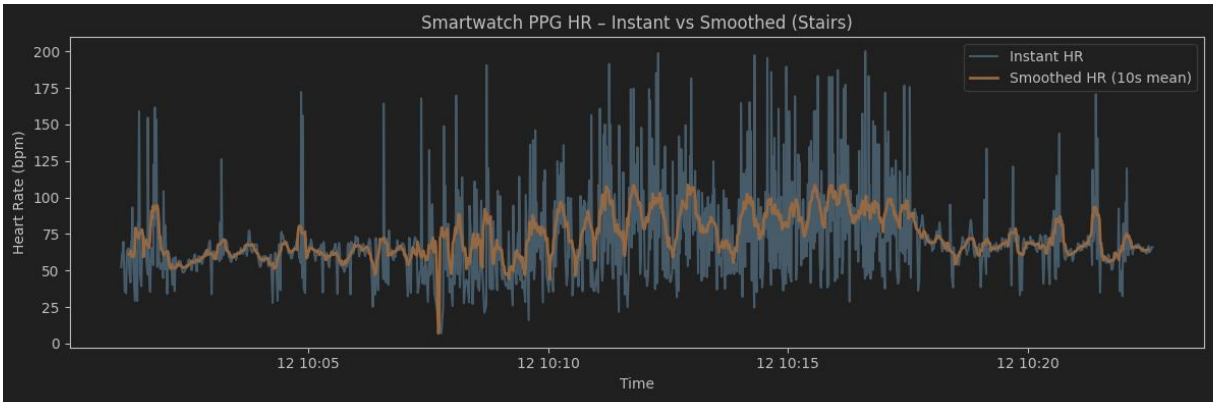


Figure 15: Heart rate during the exercise: instant heart rate (blue) and smoothed heart rate (orange)

The differences are the same as for the previous two questions. However it is notable that the subjects heart rate does not appear to rise as much as it did during this exercise when the heart rate was derived from the BioPac ECG data, and the smoothed heart rate is also much more prone to change. (Compare Figure 9)

Q6 Consider the data "Stairs" and use the biopac PPG data. Document which one you are using in order to allocate the data to the correct tracks above. Use the neurokit2 toolbox to perform PPG peak detection.
Then calculate the heart rate for each detected beat without (!) the help of the toolbox.
Calculate in 10s windows sliding it by 1s the mean heart rate and assign the calculated value to the middle of the window. That way you have one heart rate per second starting at 5s.
Use the same segments as in Q5:
Plot a 20s segment of the PPG with the corresponding beat detections, where the detection worked well.
Plot a 20s segment of the PPG with the corresponding beat detections, where the detection was faulty.
Plot the course of the instantaneous heart rate over time together with the course of the smoothed heart rate where you used sliding windows.
Do you notice any differences between the courses in Q5 and Q6?

TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

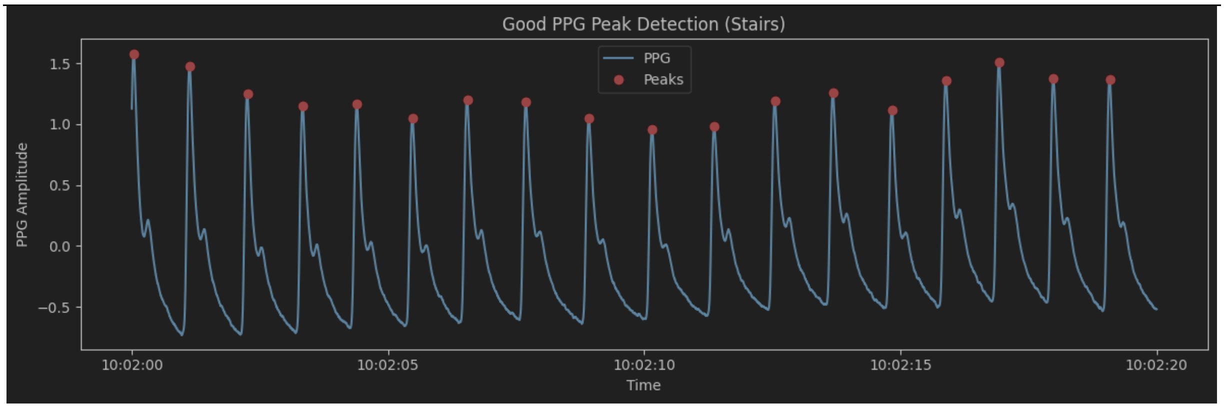


Figure 16: R-Peak Detection that worked well on BioPac PPG

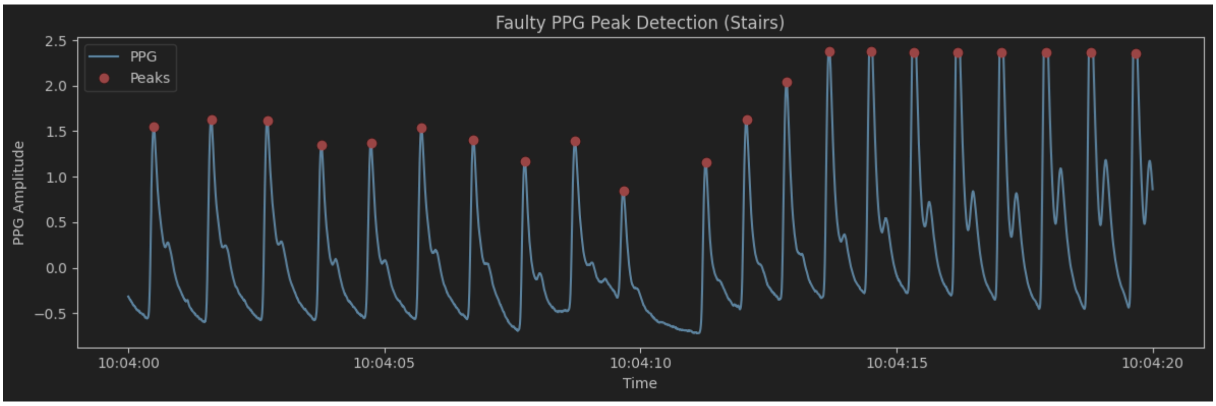


Figure 17: R-Peak Detection that worked poorly on BioPac PPG (Note: we were unable to find a segment where it worked poorly)

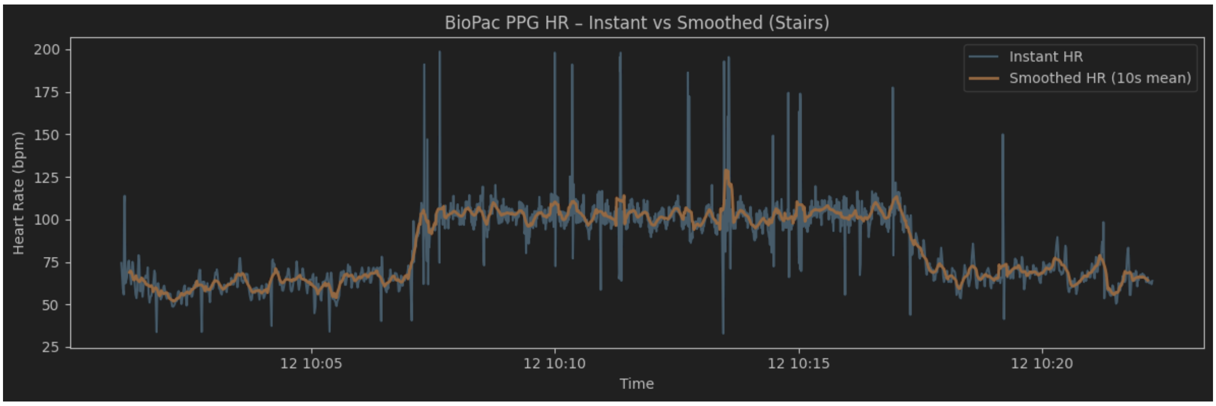


Figure 18: Heart rate during the exercise: instant heart rate (blue) and smoothed heart rate (orange)

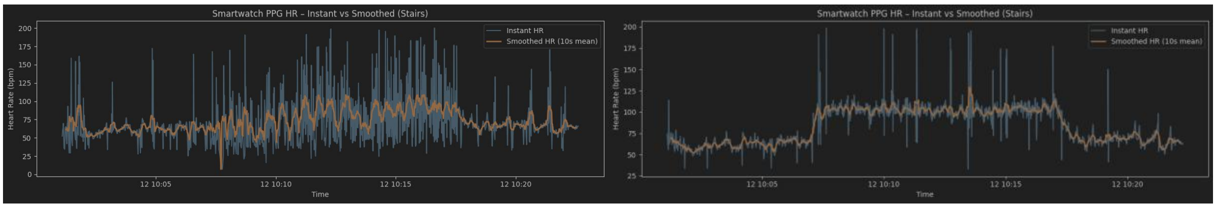


Figure 19: Figure 15 (left) and Figure 18 (right) side-by-side to aid comparison

The PPG taken as measured by the BioPac, on a small scale, mirrors the idealized PPG curve much more closely than the PPG measured by the smartwatch, where only the systole and (for most instances) the

TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

diastole are clearly defined. It's also much less prone to noise. The instant heartrate derived by the BioPac less likely to show sudden jumps to extreme values, which also reflects on the smoothed heart rate, which is much less volatile. It's also possible to clearly make out the start of the exercise, the end of the exercise, and the recovery period in the BioPac data (see graph below).

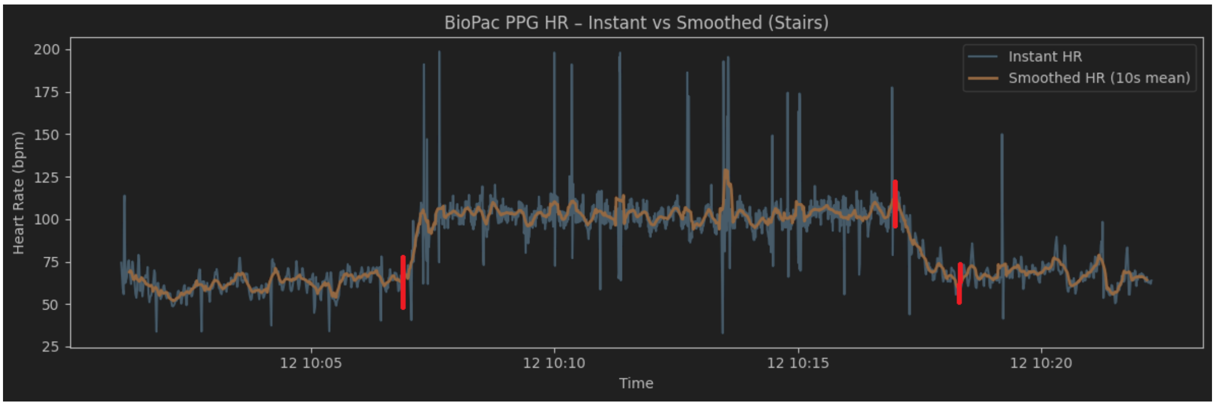


Figure 20: Manually marked beginning of exercise, end of exercise, and end of recovery period (red, left to right)

Finding the same points in the heartrate graph derived from the smartwatch PPG is much harder and less definite, as the data is much noisier. There's also nowhere near as clear a rise and fall in heartrate at the beginning and end of the exercise (see Figure 15).

Additionally we note that, even though the later questions consider the BioPac ECG data the ground truth for the heartrate, the R-Peak detection was much less faulty on the BioPac PPG data (compare Figure 10 and 11 to Figure 16 and 17), and therefore the heartrate derived from the PPG data is more accurate in this case.

Q7 Consider the smartwatch PPG and choose either "Workout" or "Stairs" data for this Question. Document which one you are using in order to allocate the data to the correct tracks above. Use the same 10s windows and sliding as before, but this time for each window identify the heart rate in the frequency domain, i.e. do a fast fourier transform for heart rate extraction. Before transforming the window into the frequency domain you should do some preprocessing for the 10 s window (e.g. highpass filtering at a meaningful frequency (e.g. using for example butterworth) or detrending). Then find a way to algorithmically detect the heart rate in the spectrum and assign the detected heart rate to the middle of the window. That way you have one heart rate per second starting at 5s. Describe how you did your preprocessing and heartrate extraction.

We considered the "Workout" data for this question.
For the preprocessing we used linear detrending to remove DC and slow drift, then manually applied a second-order Butterworth high-pass filter (cutoff at 0.5 Hz) to remove baseline wander and motion artefacts.
We then computed the FFT on each 10s window, with dominant frequency peaks identified between 0.7-3 Hz, which resulted in a typical heartrate range of approximately 42-180 beats per minute.
For the sliding window we applied a 10s window length and a 1s step size (estimate heartrate of 1Hz), then applied the heartrate to the middle of each window (meaning the first entry is 5s into the recording).

TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

	timestamp	hr
	None Count 0 531 Unique values	None Count 0
0	2025-06-12 10:42:57	42.00000
1	2025-06-12 10:42:58	42.00000
2	2025-06-12 10:42:59	78.00000
3	2025-06-12 10:43:00	48.00000
4	2025-06-12 10:43:01	48.00000
5	2025-06-12 10:43:02	48.00000
6	2025-06-12 10:43:03	42.00000
7	2025-06-12 10:43:04	48.00000
8	2025-06-12 10:43:05	54.00000
9	2025-06-12 10:43:06	42.00000
10	2025-06-12 10:43:07	42.00000
11	2025-06-12 10:43:08	42.00000
12	2025-06-12 10:43:09	42.00000
13	2025-06-12 10:43:10	42.00000
14	2025-06-12 10:43:11	66.00000
15	2025-06-12 10:43:12	66.00000
16	2025-06-12 10:43:13	66.00000
17	2025-06-12 10:43:14	66.00000
18	2025-06-12 10:43:15	84.00000
19	2025-06-12 10:43:16	72.00000

Figure 21: Excerpt from the table containing the heartrates. As the full table is 531 rows long, we refer to the code for its full scope

TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

Q8 Consider the ECG heart rate being ground truth. Calculate the mean absolute error (MAE) and the Pearson correlation coefficient (only using numpy for both!) of the smartwatch PPG HR with and without smoothing for all measurements (You have to do that for all measurements, including those that haven't been addressed in Q1- Q7 so far). Write down the equations for MAE and person correlation coefficient. Also calculate the MAE and Pearson correlation coefficient for each phase of your exercise ('walking stairs', 'resting', 12.06.2025 Task 3 – Real-world data acquisition and handling 8 'squats', 'jumping jacks', 'rotate wrist, 'circle arms'). Report the quantitative results as table. Discuss the differences that might occur (one paragraph, 15 lines max).
Repeat Q8 for the biopac PPG and discuss any differences.
Add MAE and Pearson correlation coefficient for the HR extracted from the frequency domain in your table.
Which results are the best? Discuss differences and possible reasons (one paragraph, 15 lines max).

MEASUREMENT	MAE INST HR	MAE SMOOTHED HR	PEARSON INST HR	PEARSON SMOOTHED HR
REST	21.70126	15.62513	-0.02615	-0.04666
STAIRS	29.00380	12.10610	0.07001	0.42931
WORKOUT	27.89663	16.76360	0.20511	0.46218
WORKOUT: SQUADS	56.30776	53.83409	-0.04708	0.04528
WORKOUT: STAIRS	44.37844	10.46837	-0.12719	0.03540
WORKOUT: JUMPING JACKS	38.35212	17.44166	-0.17812	-0.69327
WORKOUT: WRISTS	29.76163	15.41443	0.07348	0.70747
WORKOUT: STRETCHING	42.74303	25.16558	-0.35630	-0.72198

Table 1: Measurements of smartwatch PPG data

the smartwatch PPG data reveals large errors and low to negative Pearson correlations, especially in unsmoothed heart rate estimates. Movements such as squats, stairs, and jumping jacks cause motion artifacts that degrade signal quality. Smoothing improves both MAE and correlation substantially, showing the effectiveness of temporal aggregation. Some activities (e.g., wrist rotation) show better correlation. Negative correlations in some phases suggest inaccuracy.

MEASUREMENT	MAE INST HR	MAE SMOOTHED HR	PEARSON INST HR	PEARSON SMOOTHED HR
REST	1.21670	1.89449	0.67825	0.71795
STAIRS	16.39772	6.05780	0.35628	0.92111
WORKOUT	15.35480	6.88531	0.33562	0.68287
WORKOUT: SQUADS	41.52059	29.70983	-0.37568	-0.08026
WORKOUT: STAIRS	45.28337	23.38184	0.15122	-0.69764
WORKOUT: JUMPING JACKS	48.15474	16.71015	-0.01662	-0.65299
WORKOUT: WRISTS	35.45509	19.88176	0.10558	0.39743
WORKOUT: STRETCHING	34.84197	9.55421	0.07529	-0.27388

TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

Table 2: Measurements of BioPac PPG data

The Biopac PPG data shows better agreement with ECG ground truth compared to smartwatch PPG. Even before smoothing, the MAEs are lower and the correlation higher. After smoothing, values improve further, especially for the "Stairs" and "Rest" phases. Some exceptions exist (e.g., "Workout: Stairs"). Overall, the Biopac sensor demonstrates higher fidelity and robustness, particularly in controlled or stationary conditions.

MEASUREMENT	MAE FFT HR	PEARSON FFT HR
REST	9.97172	-0.02615
STAIRS	31.69900	0.07001
WORKOUT	34.34222	0.20511
WORKOUT: SQUADS	46.66926	-0.04708
WORKOUT: STAIRS	46.54085	-0.12719
WORKOUT: JUMPING JACKS	69.02524	-0.17812
WORKOUT: WRISTS	13.16511	0.07348
WORKOUT: STRETCHING	26.90391	-0.35630

Table 3: Measurements on FFT heartrate

MEASUREMENT	MAE INST HR WATCH	MAE SMOOTHED HR WATCH	MAE INST HR BP	MAE SMOOTHED HR BP	MAE FFT HR
REST	21.70126	15.62513	1.21670	1.89449	9.97172
STAIRS	29.00380	12.10610	16.39772	6.05780	31.69900
WORKOUT	27.89663	16.76360	15.35480	6.88531	34.34222
WORKOUT: SQUADS	56.30776	53.83409	41.52059	29.70983	46.66926
WORKOUT: STAIRS	44.37844	10.46837	45.28337	23.38184	46.54085
WORKOUT: JUMPING JACKS	38.35212	17.44166	48.15474	16.71015	69.02524
WORKOUT: WRISTS	29.76163	15.41443	35.45509	19.88176	13.16511
WORKOUT: STRETCHING	42.74303	25.16558	34.84197	9.55421	26.90391

Table 5: MAE values for all 5 considered heartrates, for comparison

MEASUREMENT	PEARSON INST HR WATCH	PEARSON SMOOTHED HR WATCH	PEARSON INST HR BP	PEARSON SMOOTHED HR BP	PEARSON FFT HR
REST	-0.02615	-0.04666	0.67825	0.71795	-0.02615
STAIRS	0.07001	0.42931	0.35628	0.92111	0.07001
WORKOUT	0.20511	0.46218	0.33562	0.68287	0.20511
WORKOUT: SQUADS	-0.04708	0.04528	-0.37568	-0.08026	-0.04708
WORKOUT: STAIRS	-0.12719	0.03540	0.15122	-0.69764	-0.12719
WORKOUT: JUMPING JACKS	-0.17812	-0.69327	-0.01662	-0.65299	-0.17812

TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

WORKOUT: WRISTS	0.07348	0.70747	0.10558	0.39743	0.07348
WORKOUT: STRETCHING	-0.35630	-0.72198	0.07529	-0.27388	-0.35630

Table 6: Pearson values for all 5 considered heartrates, for comparison

Comparing all HR extraction methods (smartwatch PPG, Biopac PPG, FFT), the smoothed Biopac PPG heart rate performs best — it achieves the lowest MAE and highest Pearson correlation across almost all activities. The smartwatch PPG, though improved by smoothing, still shows notable error and low correlation, particularly under dynamic conditions. The FFT-based HR performs worst overall — it is highly sensitive to signal quality, windowing effects, and lacks temporal resolution. Its results show poor alignment with ECG ground truth. In conclusion, the best method is smoothed HR from Biopac PPG, combining high accuracy, stability, and strong agreement with ECG data.

Q9 Do some research on Bland Altmann plots. Consider the ECG heart rate being ground truth. Plot the biopac PPG heart rate with the ground truth as a Bland Altmann plot for both smoothed and not smoothed heart rate and also for "Stairs" and "Workout" data (you can choose one measurement, but document which one).
Plot the smartwatch PPG heart rate with the ground truth as a Bland Altmann plot for both smoothed and not smoothed heart rate and also for "Stairs" and "Workout" data (you can choose one measurement, but document which one).
Also do a Bland Altmann plot on the frequency based heart rate extraction.
Using one of your Bland Altmann plots (you choose) explain the principle of this plot. What is seen and how can you interpret it in this context? (one paragraph, 15 lines max).
Discuss differences that might occur in the plots (one paragraph, 15 lines max).

We used the "Workout" data.

1. A Bland-Altman plot is a method to compare two measurement techniques by plotting the difference between the methods against their mean. It highlights bias (mean difference), limits of agreement, and potential trends in measurement error across the measurement range. In the Biopac PPG "Workout" smoothed data vs ECG ground truth plot, most points cluster around the bias line with random scatter, suggesting reasonable agreement but some systematic difference in estimation at the lower heart rates and some at higher heart rates. The limits of agreement indicate the expected range of differences between methods. This helps assess both accuracy (bias) and precision (spread). In this context, it shows how reliable the Biopac PPG HR Detection is at tracking HR changes during workouts.
2. Based on the Bland-Altman plots you can see that the BioPac PPG Based HR Data seems to work better than the data generated out of the smartwatch ppg curve. However, this might be due to the BP ECG Data being used as the ground truth. All in all though, the different BP and SmartWatch Plots show that the mean difference is in a lot of datapoints around 0 (while the random scatter is mostly in the positive), which suggests that the differ methods have a low bias and a relatively low spread. On the other hand, the frequency-based HR extraction seems to not work very well (seen by the extremely high differences in the last two plots), the values have almost nothing in common with the selected ground truth and therefore show that this method did not work as well as hoped.

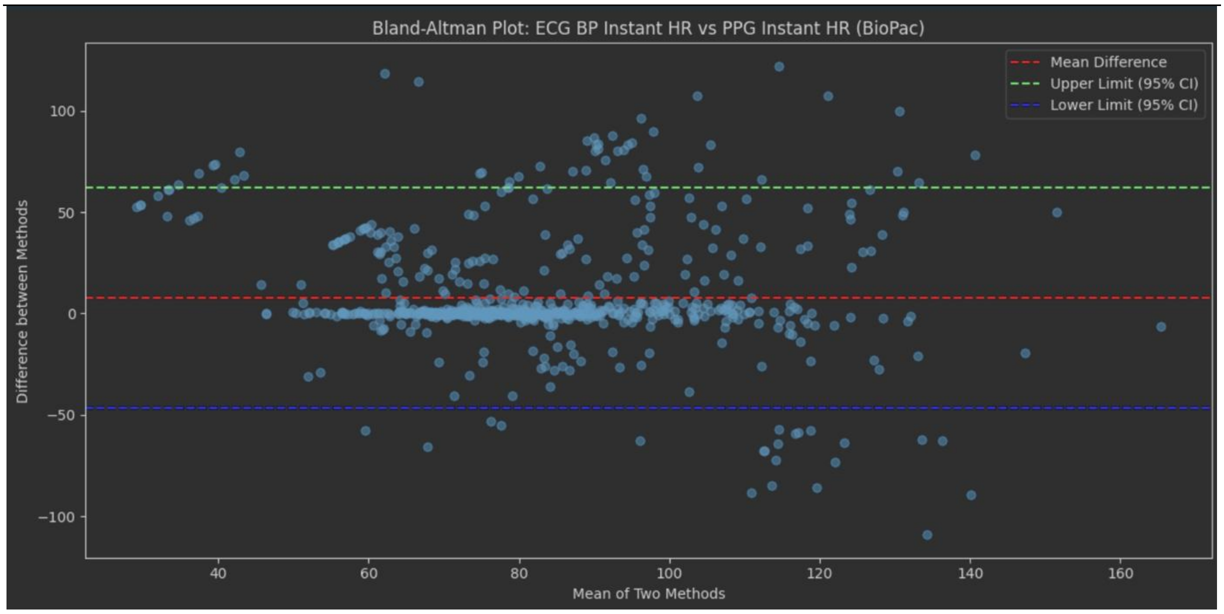


Figure 22: Bland-Altman plot of instant heartrate from BioPac ECG (ground truth) against instant heart rate from BioPac PPG

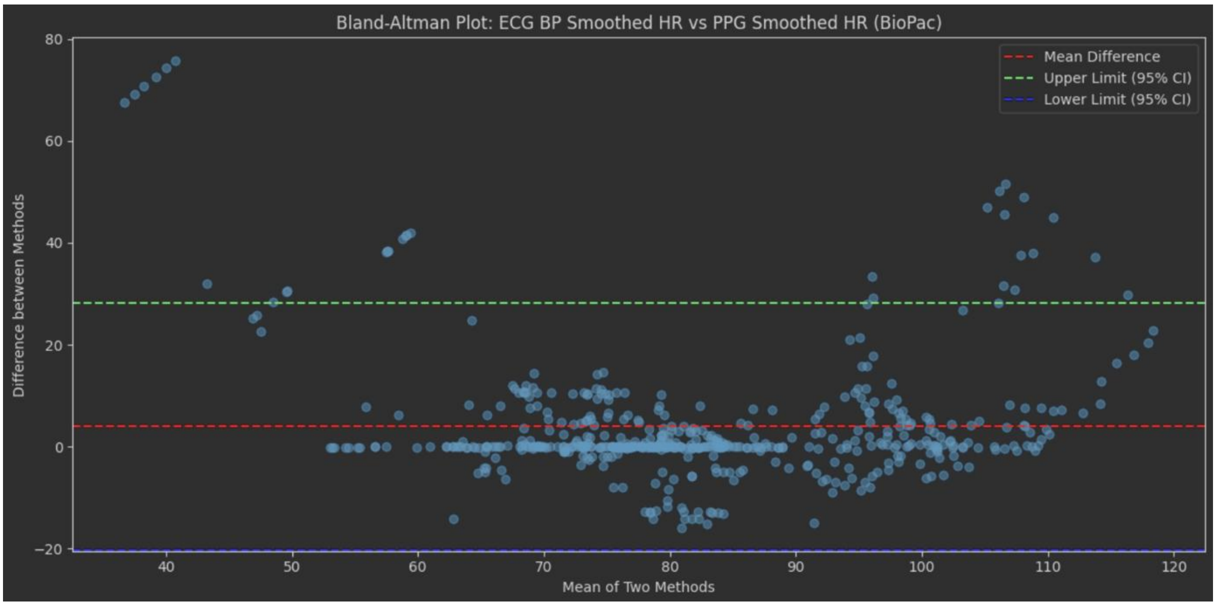


Figure 23: Bland-Altman plot of smoothed heartrate from BioPac ECG (ground truth) against smoothed heart rate from BioPac PPG

TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

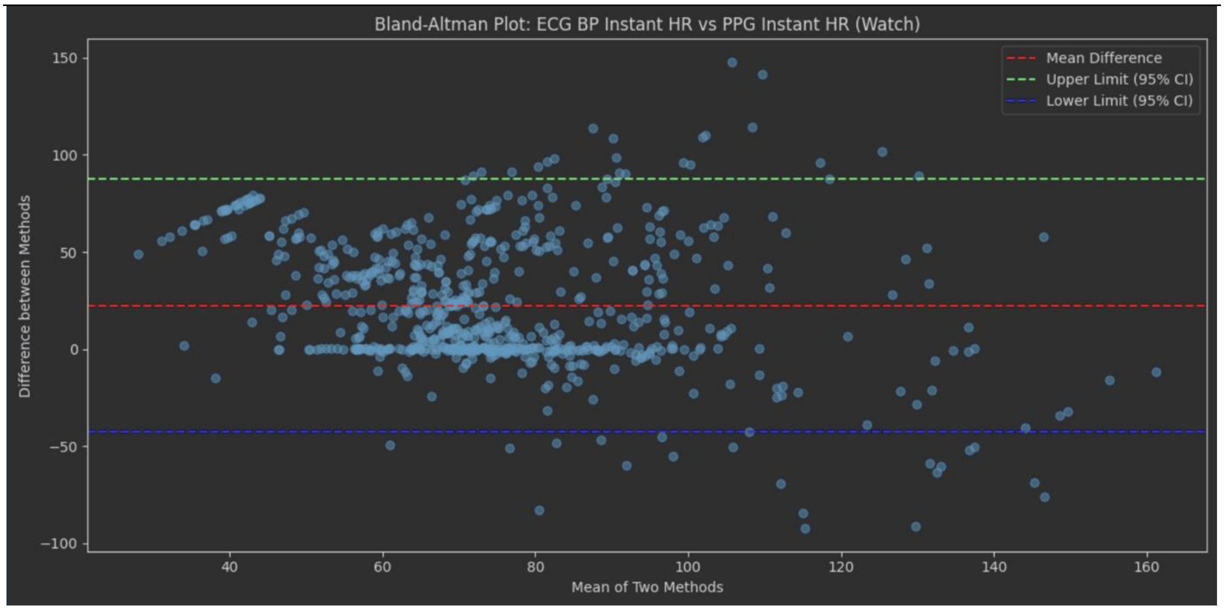


Figure 24: Bland-Altman plot of instant heartrate from BioPac ECG (ground truth) against instant heart rate from smartwatch PPG

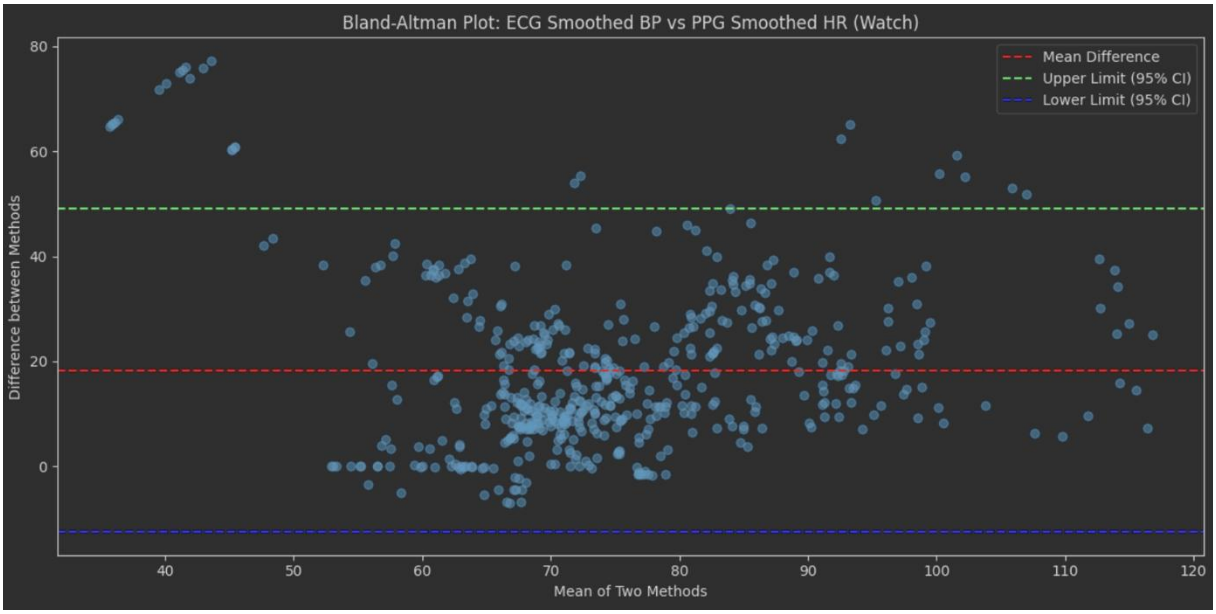


Figure 25: Bland-Altman plot of smoothed heartrate from BioPac ECG (ground truth) against smoothed heart rate from smartwatch PPG

TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

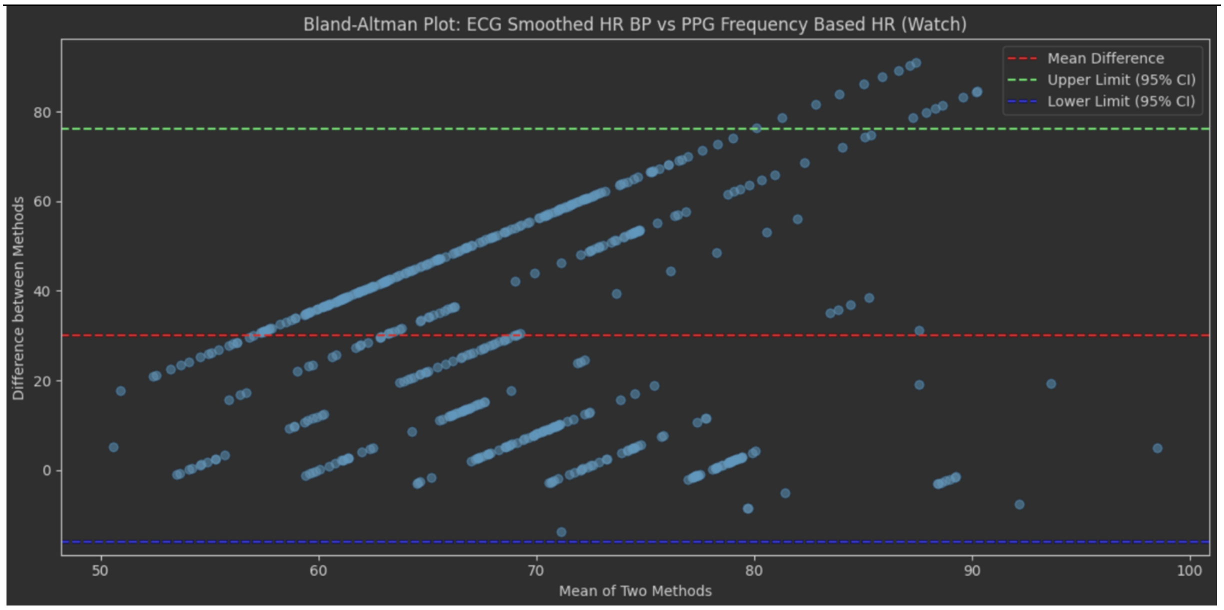
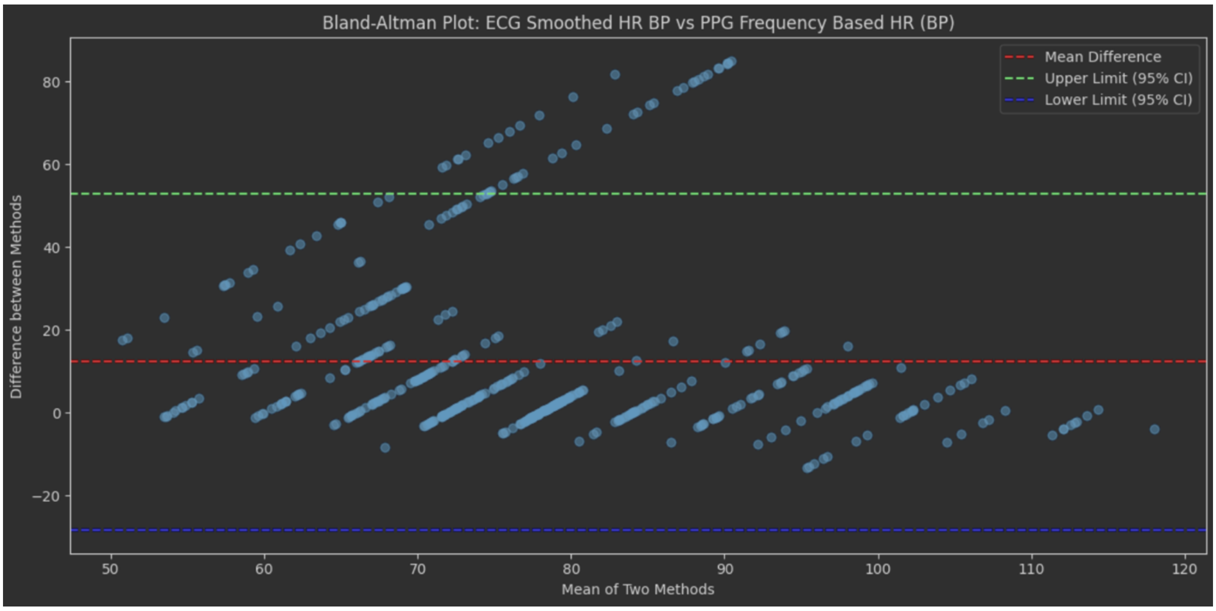


Figure 26: Bland-Altman plot of smoothed heartrate from BioPac ECG (ground truth) against frequency based heartrate calculated from smartwatch data



2 LITERATURE

[1] 02 Monitoring fundamentals.pdf (Lecture Slides)